

Matching User Preferences: Optimal Assignment of Students to Projects

Project Plan

Gabriel Pereira Viterbo

Gwendal Deléage

Jasser Mejri

Rafael Sesoko

Sébastien Rivière

Supervisor: Ekhine Irurozki Arrieta

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Note on the planning. This timeline is a preliminary, approximate version of the project plan. The spring break fell right after our first meeting with our supervisor, which limited the amount of scoping we could do collectively before the Project Plan deadline. The plan presented here may therefore be revised and complemented once we resume work after the break, in particular regarding the precise scope of the Phase 2 extensions and the allocation of tasks among group members.

1 Project Description

The matching problem we tackle is the assignment of students to projects, where each student submits an ordinal preference list over the available projects and the school must produce a final assignment that is satisfactory and fair. For the latest DSAI project assignment, this allocation was solved using a Linear Program. Now, our goal is to implement and compare classical algorithms, evaluate them on the same dataset and explore extensions that go beyond the standard one-shot, one-sided formulation.

Our main theoretical groundwork is Chapter 14 of the *Handbook of Computational Social Choice* (Klaus, Manlove, Rossi), which classifies bipartite matching problems with preferences into one-sided (House Allocation, HA) and two-sided (Hospitals/Residents, HR) variants and presents the canonical algorithms.

1.1 Tasks

Phase 1: Classical mechanisms (weeks 1–5, up to the midterm). Read Chapter 14 and implement the three canonical algorithms covered there: Serial Dictatorship (**SD**), Gale-Shapley deferred acceptance (**GS**, both applicant and proposer-oriented variants), and Top Trading Cycles (**TTC**). Define the evaluation metrics (stability function and cost function): a *cost function* aggregating individual ranks (e.g. sum of obtained ranks, number of unmatched students, Borda welfare), and a *stability score* counting blocking pairs when one side has priorities (relevant when the school ranks students by GPA). Run all three mechanisms on the course data and on synthetic instances. The midterm evaluation on May 29 will be the deadline for this phase.

Phase 2: Extensions and analysis (weeks 6–8). Once the baselines are in place, we explore variants of the problem and propose our own approaches. Candidate directions, to be narrowed down at the end of Phase 1:

- **Two-sided preferences with priorities:** the school also ranks students (e.g. by GPA), turning the problem into an instance of HR; compare the student-optimal and school-optimal stable matchings.
- **Partial preferences and ties:** truncated preference lists and indifferences (HRT-style instances), with appropriate tie-breaking rules.
- **Veto**es and **constraints:** forbidden student–project pairs, project quotas, minimum group sizes.
- **Coalitions and groups:** students who must be assigned together, motivated by team projects.
- **Statistical setting:** treating reported preferences as noisy samples from an underlying probability distribution

Phase 3: Synthesis (week 9). Final report and oral presentation by the June 26 deadline: consolidate results, articulate trade-offs, and give a recommendation for which mechanism (or combination) we believe best fits the Télécom Paris use case.

1.2 Data

Two data sources will be used in parallel:

- **Real data** collected within the latest DSAI Project matching: anonymised preference lists submitted by students over the available projects.
- **Synthetic data** generated under controlled preference models

2 Deliverables

- Git repository with the implementations of SD, GS, TTC, and the Phase 2 extensions.
- Reproducible experimental notebooks comparing the mechanisms on the course data and on synthetic benchmarks.
- Final report describing the methods and results
- Oral presentation of the project.

3 Task Allocation

The tables below summarise the distribution of responsibilities among the group members for each phase of the project. This allocation is a proposal to be refined as the project progresses, in particular after the midterm evaluation, based on the Phase 1 results and on each member’s interests. All members will contribute to the writing of the reports and to the preparation of the oral presentations.

3.1 Phase 1 – Classical mechanisms (weeks 1–5)

Member	Main responsibility	Tasks
Gabriel Viterbo	Serial Dictatorship	Implementation of Serial Dictatorship (SD) and its variants (random order, priority-based order); setup of the Git repository and project structure; writing the introduction and SD section of the midterm report; preparing the introduction of the midterm presentation.
Gwendal Deléage	Gale-Shapley	Implementation of Gale-Shapley (applicant-proposing and project-proposing variants); unit tests and validation on small cases; writing the Gale-Shapley section of the midterm report; preparing the Gale-Shapley part of the midterm presentation.
Rafael Sesoko	Top Trading Cycles	Implementation of Top Trading Cycles (TTC) with explicit cycle detection; unit tests and edge cases; writing the TTC section of the midterm report; preparing the TTC part of the midterm presentation.
Jasser Mejri	Evaluation metrics	Definition and implementation of the cost function; definition and implementation of the stability score; shared evaluation module used by all algorithms; writing the evaluation methodology section of the midterm report; preparing the metrics part of the midterm presentation.
Sébastien Rivière	Data & experimentation	Preparation and anonymisation of the real DSAI project data; generation of synthetic data under controlled preference models; setup of the experimental notebooks comparing the three mechanisms; writing the data and experimental results section of the midterm report; preparing the experimental results part of the midterm presentation.

Table 1: Phase 1 – Classical mechanisms (weeks 1–5)

3.2 Phase 2 – Extensions and analysis (weeks 6–8)

Each member takes ownership of one extension, building on the modules developed in Phase 1, and is responsible for writing the corresponding section of the final report and for presenting it during the final oral.

Member	Extension	Tasks
Gabriel Viterbo	Two-sided preferences with priorities (HR)	Extending the problem to Hospitals/Residents (the school also ranks students, e.g. by GPA); comparison of student-optimal and school-optimal stable matchings; coordination of the final integration and overall structure of the final report.
Gwendal Deléage	Partial preferences and ties (HRT)	Truncated preference lists and indifference; implementation and comparison of different tie-breaking rules; analysis of the impact on stability and cost.
Jasser Mejri	Veto and constraints	Forbidden student–project pairs; project quotas and minimum group sizes; adaptation of the three Phase 1 algorithms to handle these constraints.
Rafael Sesoko	Coalitions and groups	Students who must be assigned together (motivated by team projects); adaptation of the mechanisms to handle coalitions; analysis of the trade-offs between individual satisfaction and group constraints.
Sébastien Rivière	Statistical setting	Modelling reported preferences as noisy samples from an underlying distribution; experiments on synthetic data generated under this model; robustness analysis of the mechanisms with respect to noise in preferences.

Table 2: Phase 2 – Extensions and analysis (weeks 6–8)

3.3 Phase 3 – Synthesis (week 9)

The final report and oral presentation (deadline 26 June) are prepared collectively. Each member contributes the section corresponding to their extension, and all members take part in writing the cross-cutting parts of the report (comparative analysis, conclusion, recommendation for Télécom Paris). The oral presentation is delivered jointly, with each member presenting their own contribution.